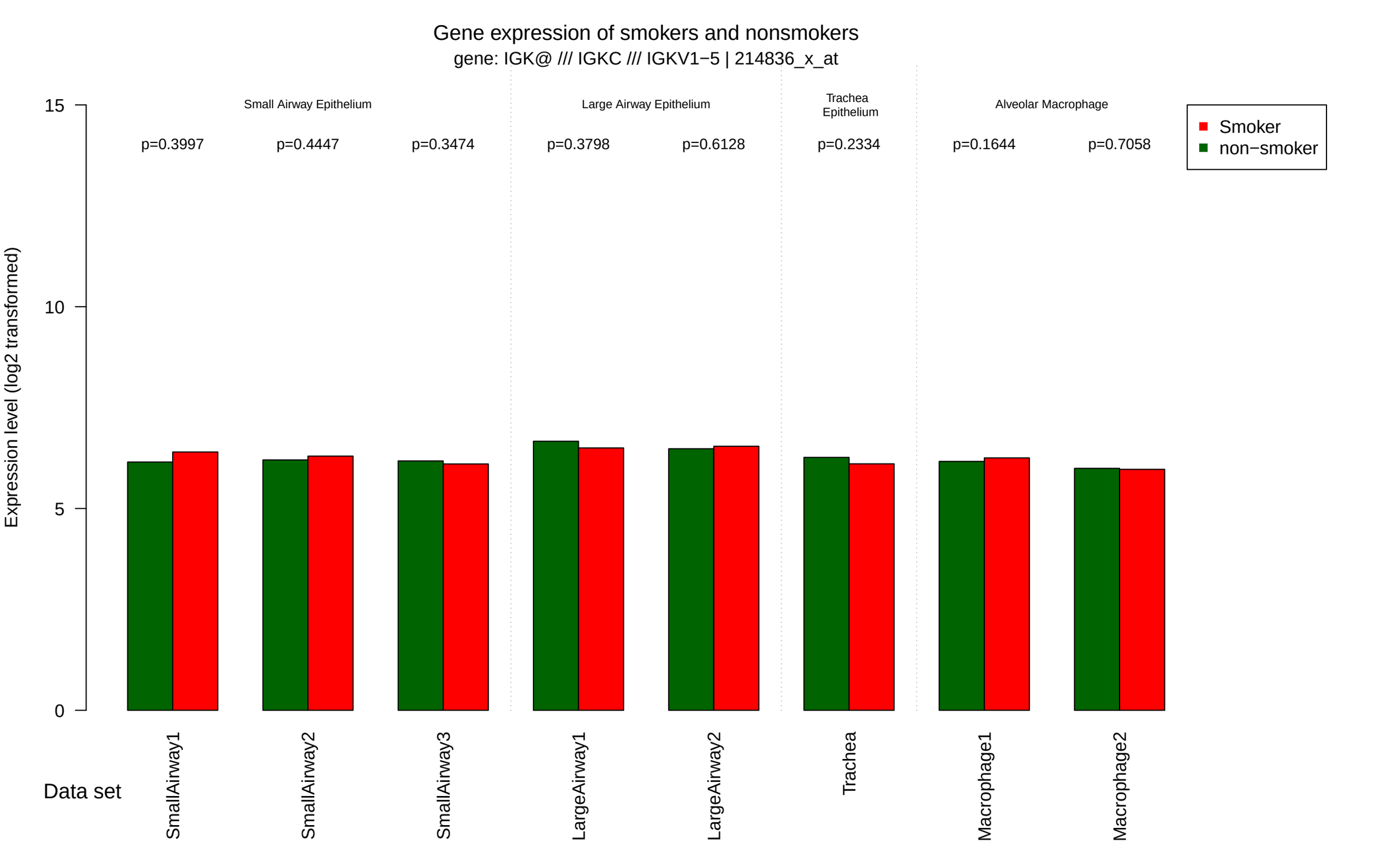
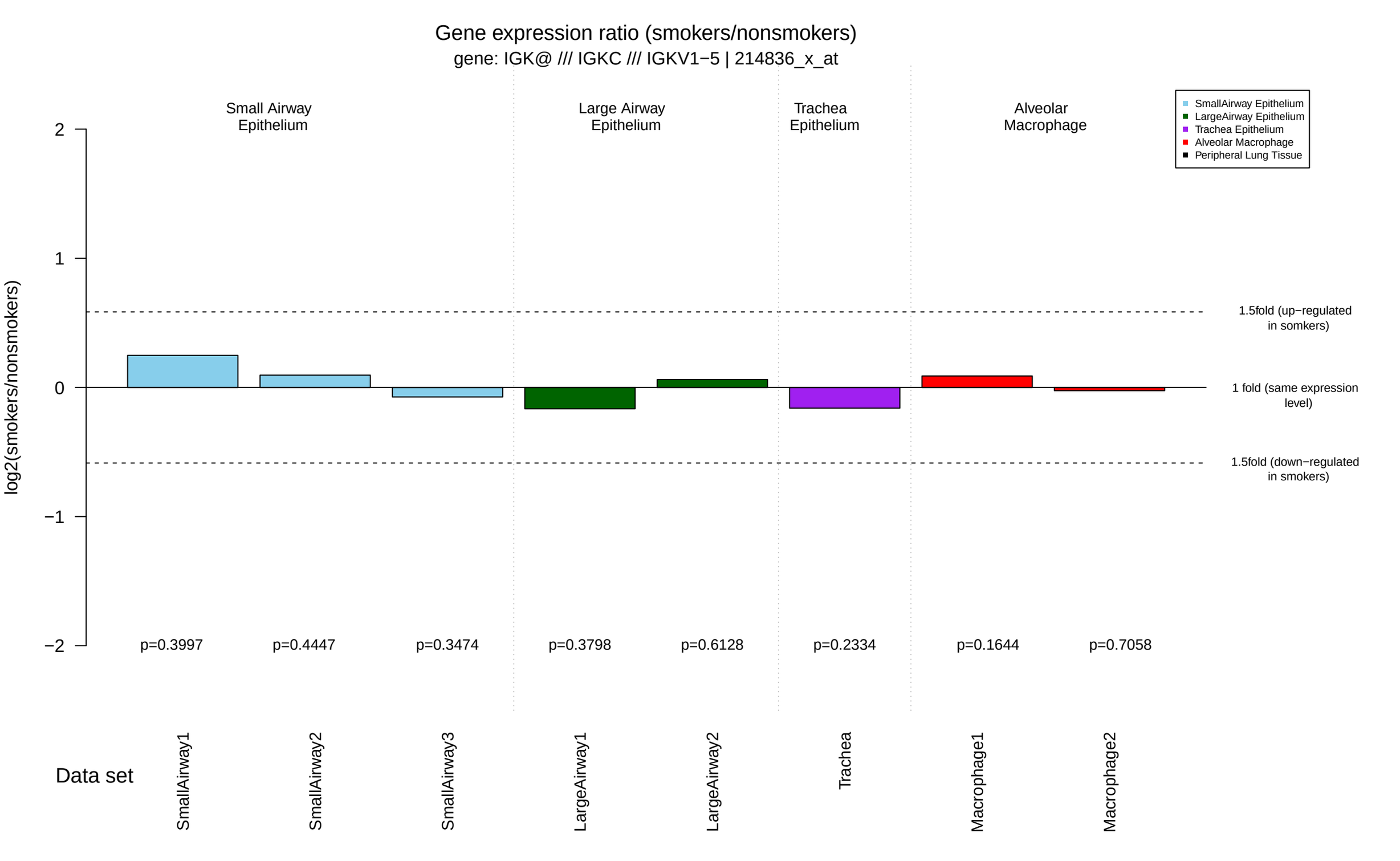
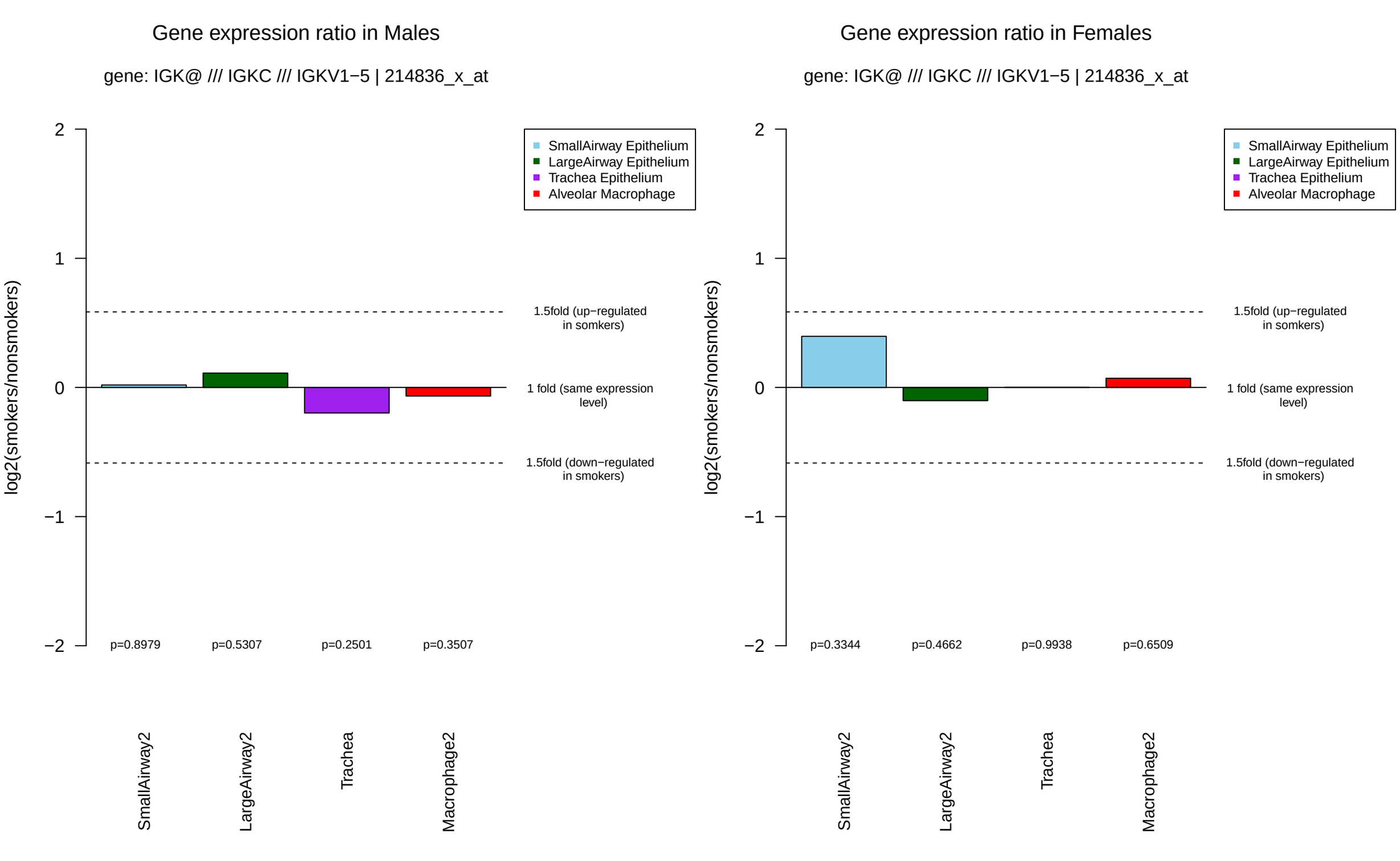
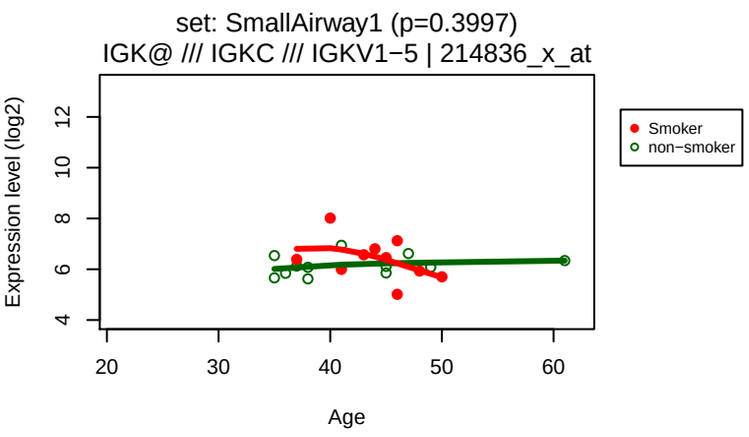
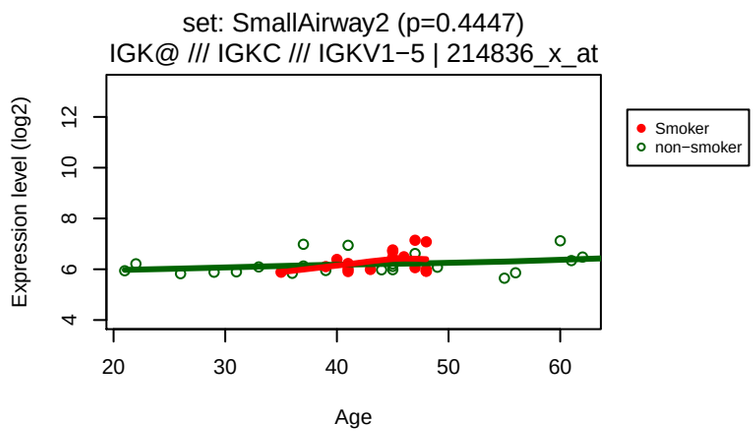
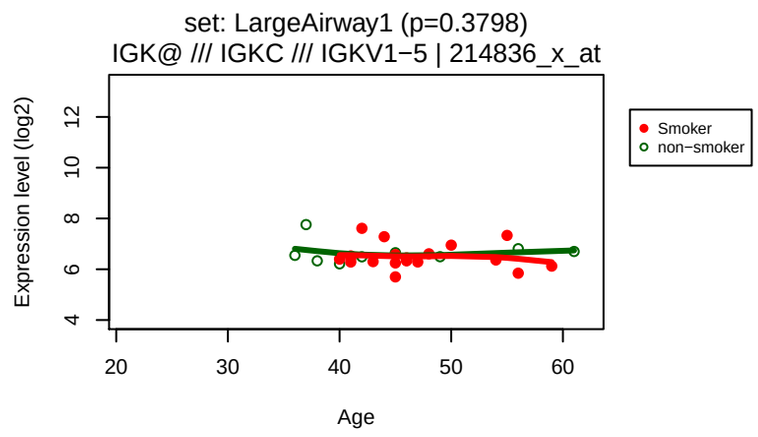

Gene expression of smokers and nonsmokers
gene: IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3997
p=0.4447
p=0.3474
p=0.3798
p=0.6128
p=0.2334
p=0.1644
p=0.7058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

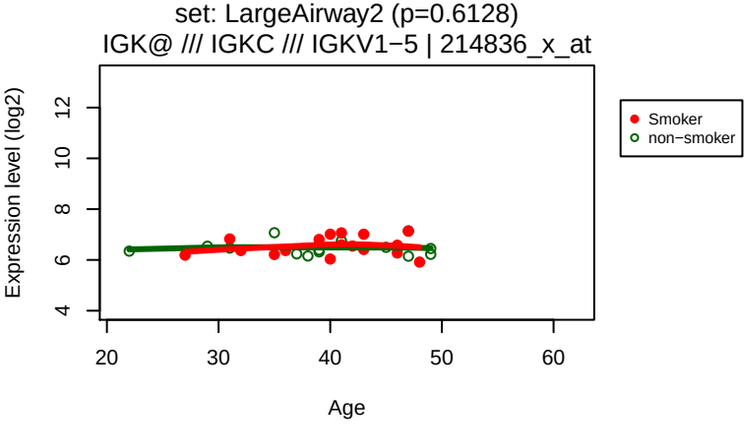

Gene expression ratio (smokers/nonsmokers)
gene: IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3997
p=0.4447
p=0.3474
p=0.3798
p=0.6128
p=0.2334
p=0.1644
p=0.7058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

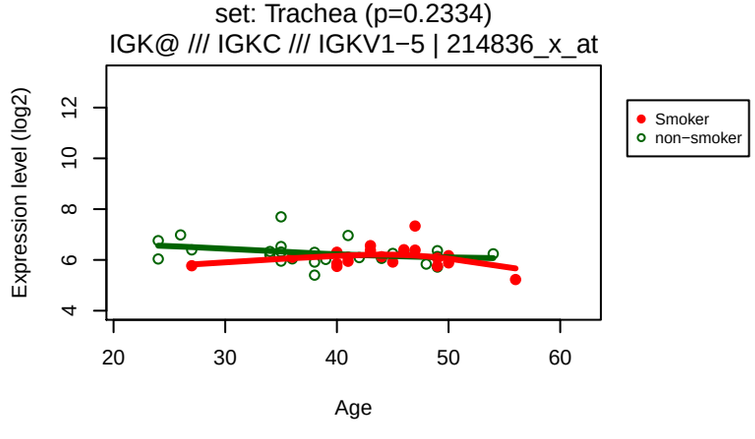

Gene expression ratio in Males
gene: IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8979
p=0.5307
p=0.2501
p=0.3507
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3344
p=0.4662
p=0.9938
p=0.6509
SmallAirway2
LargeAirway2
Trachea
Macrophage2

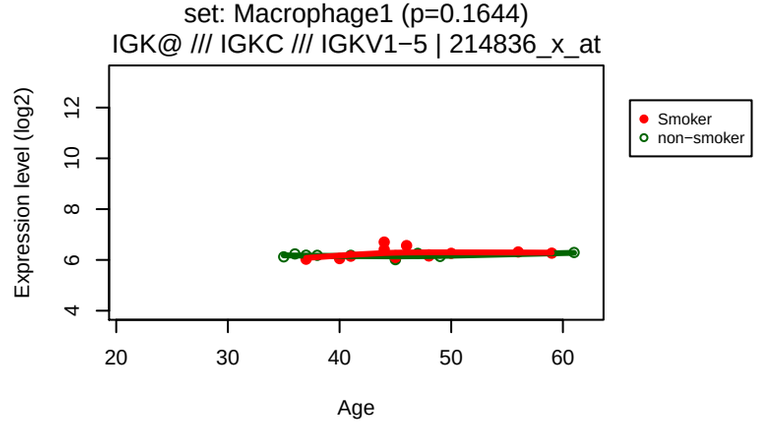

set: SmallAirway1 (p=0.3997)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker

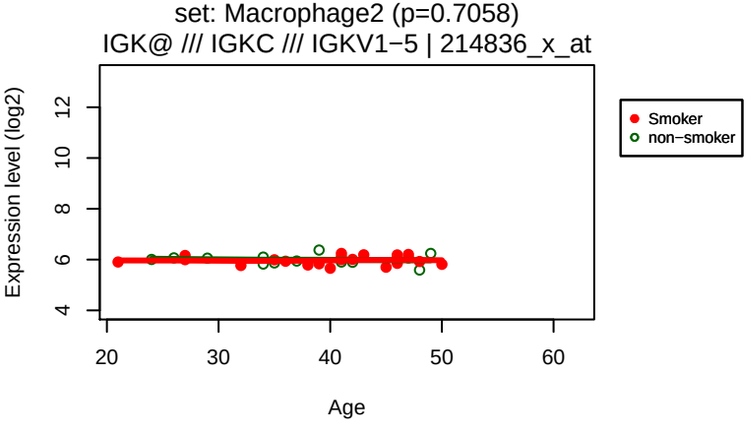

set: SmallAirway2 (p=0.4447)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker

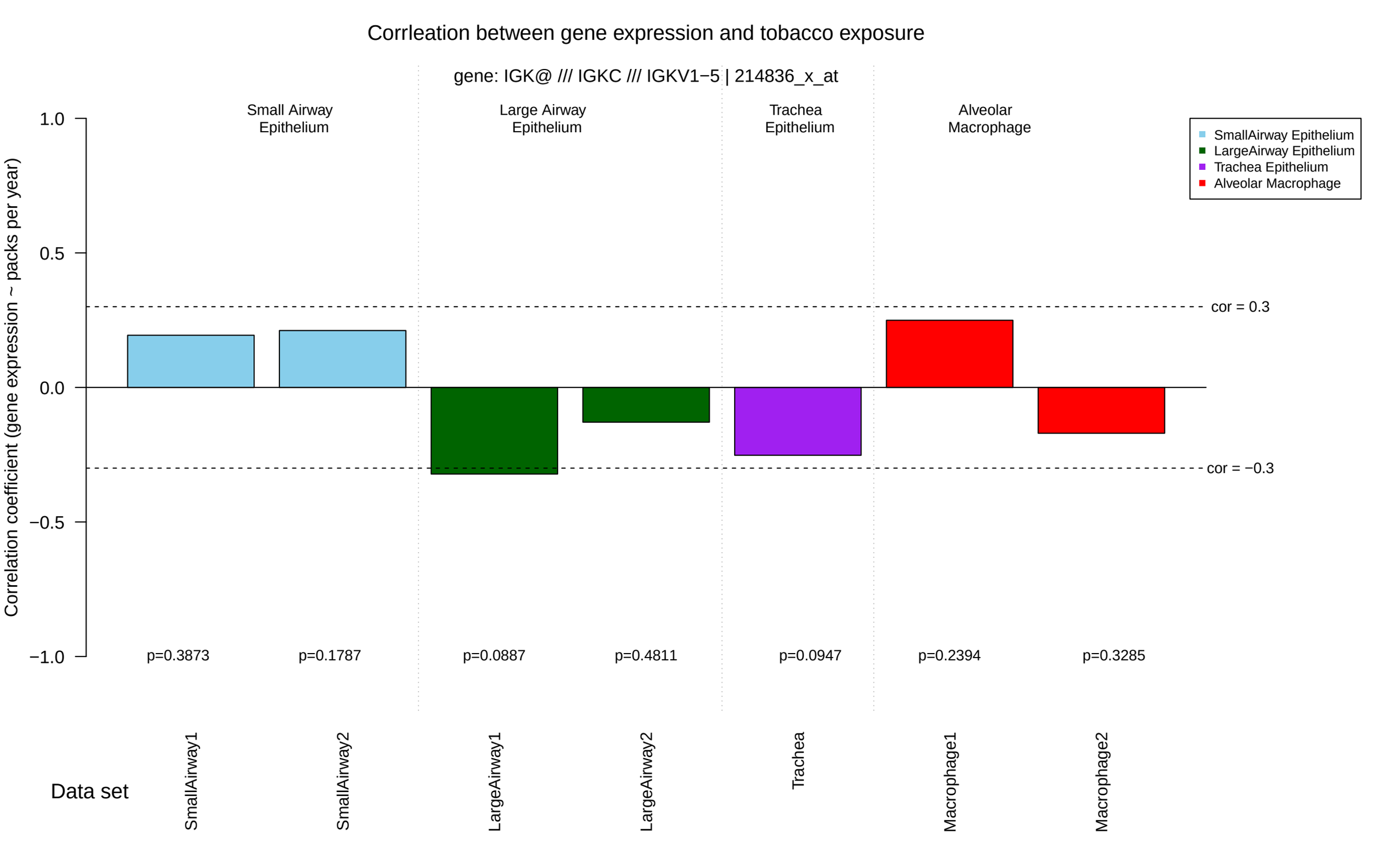

set: LargeAirway1 (p=0.3798)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6128)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2334)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1644)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7058)
IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGK@ /// IGKC /// IGKV1−5 | 214836_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3873
p=0.1787
p=0.0887
p=0.4811
p=0.0947
p=0.2394
p=0.3285
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2